Joule heating and $\mathrm{Im}(\chi)$

Recall how retarded susceptibility appears

$$\underbrace{\langle \hat{A}(\vec{r}) \rangle_t}_{\text{expectation value of } \hat{A}} = -\frac{i}{\hbar} \int_{-\infty}^{t} dt' \int d\vec{r}' \, \langle [\hat{A}(r,t), \hat{O}(r',t')] \rangle_0 \, h(\vec{r}',t')$$

in response to t-dependent perturbation $H'(t) = \int d\vec{r} \, h(\vec{r},t) \, \hat{O}(r)$

For the density-density susceptibility

$$\chi_0(\vec{r},t) = -i\,\theta(t) \langle [\hat{\rho}(r,t), \hat{\rho}(0)] \rangle$$

we have $\hat{A} = \hat{O} = \hat{\rho}$.

Perturbation $H'(t) = \int d\vec{r} \, \hat{\rho}(r,t) \, \underbrace{\varphi(r,t)}_{\text{(electrical) potential}}$

That creates (longitudinal) electric field

$$\vec{E}(r,t) = -\vec{\nabla} \varphi(r,t) \Rightarrow \vec{E}(\vec{q},\omega) = -i\,\vec{q}\,\varphi(\vec{q},\omega)$$

and electrical current $\vec{J}(r,t) = \sigma \vec{E}$ ← relation maybe nonlocal in $\vec{r}$-space but

$$\vec{J}(\vec{q},\omega) = \sigma(\vec{q},\omega) \, \vec{E}(\vec{q},\omega) \quad \text{(Ohm's law)}$$

We now find relation between σ and χ:

from continuity equation we know that

$$\partial_t \rho + \vec{\nabla} \cdot \vec{J} = 0 \Rightarrow -i\omega \rho(\vec{q},\omega) + i\,\vec{q} \cdot \vec{J}(\vec{q},\omega) = 0$$

$$\Rightarrow \vec{q}\cdot\vec{J}(\vec{q},\omega) = \sigma(\vec{q},\omega)\,\vec{q}\cdot\vec{E}(\vec{q},\omega) =$$

$$= \omega\,\rho(\vec{q},\omega) = \omega\left\{\chi(\vec{q},\omega)\,\varphi(\vec{q},\omega)\right\}$$

density appears in response to φ

But $\vec{q}\cdot\vec{E}(\vec{q},\omega) = \vec{q}\cdot\left(-i\vec{q}\,\varphi(\vec{q},\omega)\right) = -i\,q^2\,\varphi(\vec{q},\omega)$

$$\Rightarrow -i\,\sigma(\vec{q},\omega)\,q^2\,\varphi(\vec{q},\omega) = \omega\,\chi(\vec{q},\omega)\,\varphi(\vec{q},\omega)$$

Longitudinal conductivity $\quad \sigma(\vec{q},\omega) = \dfrac{i\omega}{q^2}\,\chi(\vec{q},\omega) \quad$ density-density susceptibility

$$\Rightarrow \mathrm{Re}\left[\sigma(\vec{q},\omega)\right] = -\frac{\omega}{q^2}\,\mathrm{Im}\left[\chi(\vec{q},\omega)\right]$$

$$\left(\text{because} \quad \mathrm{Re}\left(\frac{i\omega}{q^2}\chi(\vec{q},\omega)\right) = \mathrm{Re}\left(\frac{i\omega}{q^2}(\chi' + i\chi'')\right)\right).$$

We found previously (see posted lecture notes) that generally the rate of energy absorption is given by

$$\overline{\frac{dE}{dt}} = -2\,\omega\,\mathrm{Im}\left(\chi(\vec{q},\omega)\right)\,|\varphi(\vec{q},\omega)|^2 =$$

$$= 2\,q^2\,\mathrm{Re}\left(\sigma(\vec{q},\omega)\right)\,\underbrace{\varphi^*(\vec{q},\omega)}_{\left(\frac{\vec{E}(\vec{q},\omega)}{-i\vec{q}}\right)^*}\cdot\underbrace{\varphi(\vec{q},\omega)}_{\frac{\vec{E}(\vec{q},\omega)}{-i\vec{q}}}$$

$$= 2\,\mathrm{Re}\left[\sigma(\vec{q},\omega)\right]\,\left|\vec{E}(\vec{q},\omega)\right|^2$$

This is just Ohmic heat!

So we apply monochromatic $\vec{E}(t) = \vec{E}_0(q,\omega)\, e^{i(\vec{q}\cdot\vec{r}-\omega t)} + c.c.$
(complex conjugate)

It drives current $\vec{J}(r,t) = \vec{J}_0(q,\omega)\, e^{i(\vec{q}\cdot\vec{r}-\omega t)} + c.c.$

Power dissipated at point $\vec{r}$, time t is

$$P(\vec{r},t) = \vec{J}\cdot\vec{E} = \left(\vec{J}_0(\bar{q},\omega)\, e^{i(\bar{q}\cdot\bar{r}-\omega t)} + \vec{J}_0^*(\bar{q},\omega)\, e^{-i(\bar{q}\cdot\bar{r}-\omega t)}\right)$$
$$\cdot\left(\vec{E}_0(\bar{q},\omega)\, e^{i(\bar{q}\cdot\bar{r}-\omega t)} + \vec{E}_0^*(\bar{q},\omega)\, e^{-i(\bar{q}\cdot\bar{r}-\omega t)}\right)$$

$$= \vec{J}_0(\bar{q},\omega)\cdot\vec{E}_0^*(\bar{q},\omega) + \vec{J}_0^*(\bar{q},\omega)\cdot\vec{E}_0(\bar{q},\omega)$$
$$+ \vec{J}_0\cdot\vec{E}_0\, e^{i2(\bar{q}\cdot\bar{r}-\omega t)} + \vec{J}_0^*\cdot\vec{E}_0^*\, e^{-i2(\bar{q}\cdot\bar{r}-\omega t)}$$

reversible terms
(describe energy going <u>in</u> and <u>out</u>)

absorption $\overline{\dfrac{dE(\vec{r})}{dt}} = \dfrac{1}{\tau}\displaystyle\int_0^{\tau=2\pi/\omega} dt\, P(\vec{r},t) \Rightarrow \int_0^{2\pi/\omega} dt\, e^{\pm i2\omega t} =$

$$= \frac{e^{\pm i2\omega t}\Big|_0^{2\pi/\omega}}{\pm i2\omega} = \frac{e^{i4\pi}-1}{\pm i2\omega} = 0.$$

$$\Rightarrow \overline{\frac{dE}{dt}} = \vec{J}_0(\bar{q},\omega)\cdot\vec{E}_0^*(\bar{q},\omega) + \vec{J}_0^*(\bar{q},\omega)\cdot\vec{E}_0(\bar{q},\omega)$$

By Ohm's law $\vec{J}_0(\bar{q},\omega) = \sigma(\bar{q},\omega)\,\vec{E}_0(\bar{q},\omega)$

$$\vec{J}_0^*(\bar{q},\omega) = \sigma^*(\bar{q},\omega)\,\vec{E}_0^*(\bar{q},\omega)$$

$$\Rightarrow \overline{\frac{dE}{dt}} = \sigma(\bar{q},\omega)\,\vec{E}_0(\bar{q},\omega)\cdot\vec{E}_0^*(\bar{q},\omega) + \sigma^*(\bar{q},\omega)\,|\vec{E}_0|^2 =$$
$$= 2\,\mathrm{Re}[\sigma(\bar{q},\omega)]\,|\vec{E}_0(\bar{q},\omega)|^2.$$

OK

Thus, we are interested in response to uniform ($\vec{q}=0$) AC electric field. Hence need the limit $\omega \gg v_F q \to 0$:

$$\chi''(\vec{q},\omega) = -\pi \nu_0 \frac{\omega}{2 v_F q} \theta\left(1 - \frac{|\omega|}{v_F q}\right) \Rightarrow 0$$

$$\chi'(\vec{q},\omega) = -\nu_0 \left(1 + \frac{\omega}{2 v_F q} \ln\left|\frac{\omega - v_F q}{\omega + v_F q}\right|\right) =$$

$$= -\nu_0 \left\{1 + \frac{\omega}{2 v_F q} \ln\left(\frac{1 - \frac{v_F q}{\omega}}{1 + \frac{v_F q}{\omega}}\right)\right\}$$

$$\Rightarrow -\nu_0 \left\{ -\frac{1}{3}\left(\frac{v_F q}{\omega}\right)^2 - \frac{1}{5}\left(\frac{v_F q}{\omega}\right)^4 \right\}$$

$$= \frac{\nu_0}{3} \frac{v_F^2 q^2}{\omega^2}\left(1 + \frac{3}{5}\frac{v_F^2 q^2}{\omega^2}\right).$$

Then $\sigma(\vec{q},\omega) = \frac{i\omega}{q^2}\chi(\vec{q},\omega) = (\vec{q}\to 0)$

$$= \frac{i\omega}{q^2}\frac{\nu_0}{3}\frac{v_F^2 q^2}{\omega^2} =$$

$$= \frac{\nu_0 v_F^2}{3}\frac{i}{\omega}$$

Thus, for ideal Fermi gas $\sigma(0,\omega) \sim \frac{i}{\omega}$ is purely imaginary $\Rightarrow$ no energy absorption.

To get it, need impurities or phonons, for then (τ = relaxation time)

$$\sigma(\omega) \sim \frac{i}{\omega + i/\tau} = \frac{i(\omega - i/\tau)}{\omega^2 + 1/\tau^2} \Rightarrow \mathrm{Re}(\sigma) \sim \frac{1/\tau}{\omega^2 + 1/\tau^2} \sim \tau$$